

Test Certificate

Product Code:	NMDC1DPB144BK
Description:	Up to 144f, 12 position Stranded loose tube, PE/NY/GC/PE
Cable #:	4152960
Date Cert created:	20 th Aug 2024
Design Number:	83 - 07
Product Family:	NMDCxxPB ⁺⁺⁺ XX

TEST PARAMETER / METHOD	AFL ACCEPTANCE CRITERIA	ACHIEVED	RESULT
Tensile Strength IEC 60794-1-21 (E1)	7 kN 30min, < 0.6% fibre strain, < 0.1dB during test, @ 1550nm No att. Increase after test,	>7kN <0.6% fs $\Delta\alpha \leq 0.1\text{dB}$ During test No att. Increase after test	Pass
Bend under tension IEC 60794-1-21 (E18A)	$r_b = 20 \times \text{OD}$ 360°, 1 turn, 1 min No fibre break, no damage to sheath or core structure and No attn. change throughout test	$r_b = < 20 \times \text{OD}_{\text{measured}}$ No breaks or damage and no attenuation change	Pass
Crush (short term) IEC 60794-1-21 (E3)	2.2kN/100mm Duration: 10min 3 positions No damage to sheath or core structure and No att. Increase after test	>2.2kN/100mm No damage and no attenuation increase	Pass
Crush (long term) IEC 60794-1-21 (E3)	2kN/100mm Duration: 120min 3 positions No damage to sheath or core structure and No att. Increase during or after test	2kN/100mm No damage and no attenuation increase	Pass
Impact IEC 60794-1-21 (E4)	Weight: 1.5.0kg Height: 1.0m , 300mm rad anvil, 3 impacts, After 5 mins No fibre break, no damage to sheath or core structure and No attn. change throughout test	Weight: 2.0kg Height: 1.0m , 10 impacts, No breaks or damage and no attenuation change	Pass
Bend IEC 60794-1-21 (E11)	$r_b = 10 \times \text{OD}$ 360° loop, 1 turn, 1 cycle No att. Change throughout test	$r_b = 10 \times \text{OD}$ (actual) No attn. Change	Pass
Torsion IEC 60794-1-21-(E7)	Sample length: 2 m Rotation: a) 180° clockwise, b) return to starting position, c) 180° anticlockwise, d) return to starting position. Four movements = one cycle. Complete 5 cycles (a to d) in one minute maximum During the tenth final cycle at a), c) and after completion (no rotation) check transmitting fibres. No fibre breaks, no damage to the sheath or to the core structure and no attenuation change throughout test	No breaks or damage and no attenuation change	Pass
Temperature Cycle IEC 60794-1-22 (F1)	Temperature Range: -40°C to +70°C @ 1550nm $\Delta\alpha_{\text{max}}$ Last cycle TA2 and TB2 $\leq 0.15\text{dB}$ $\Delta\alpha_{\text{max}}$ TA1 and TB1 $< 0.05\text{dB}$ $\Delta\alpha_{\text{max}}$ completion $< 0.05\text{dB}$	$\Delta\alpha_{\text{max}}$ Last cycle TA2 and TB2 $\leq 0.10\text{dB}$ $\Delta\alpha_{\text{max}}$ TA1 and TB1 $< 0.05\text{dB}$ $\Delta\alpha_{\text{max}}$ completion $< 0.05\text{dB}$	Pass
Water Penetration IEC 60794-1-22 (F5B)	Sample length: 3m Water Height: 1m No water after 24 hours	No Water detected	Pass

Tested By: Karlo L & Attila N	Verified By: Shashini Subasinghe	Approved By: Jim Boukouvalas
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